

Assignment 5 Prompt Log

Student: Vineesh Cutting | **Roll number:** RSI2026505

Paper: The Convergence of Artificial Intelligence and Digital Twins for Smart Farming: A Critical Review of Architectures, Applications, and Research Directions

Evidence Status

Codex prepared the local conversion and operated the student's authenticated Prism and Overleaf accounts with explicit permission on 26-27 September 2026. The six actual prompts and the approved-revision follow-up below record submitted text, responses, and decisions. Earlier unused preparation prompts are preserved separately in Evidence/prepared-prompts-historical.md. The old-topic log has not been relabelled.

Completed Session: 1 Text Conversion

Exact submitted prompt: Assignment 5 evidence task 1: text-to-LaTeX conversion. Do not edit project files. Convert the following ORIGINAL paragraph into clean LaTeX, preserving exact wording, scientific meaning, and sentence order. Return only the converted paragraph and a brief preservation check. Original: Calibration estimates model parameters, whereas data assimilation updates the estimated system state as observations arrive. Sensor calibration is a separate measurement problem. These three activities should be documented independently: an apparently successful hydraulic calibration can otherwise compensate for a biased sensor, incorrect irrigation input, or unmodelled water-table influence.

Actual response: Returned the original paragraph unchanged and stated that exact wording, meaning, and order were preserved; no escaping was needed. **Decision:** Accepted as paragraph-conversion evidence, not a claim that Prism originally converted the whole manuscript.

Completed Session: 2 Equations

Exact submitted prompt: Task 2. Read main.tex without editing files. Audit all 12 equations, their labels and symbols, especially the upward-positive Richards convention and root-extraction sign. Distinguish internal consistency from scientific validation; do not invent findings. Report concise findings.

Actual response summary: Twelve unique labels; internally consistent Richards and PINN-residual signs; worked values of 60 mm TAW, 30 mm RAW, 24 mm depletion, 20 mm gross irrigation, and 200 m3. Minor vector/matrix typography inconsistencies and reused symbols were noted. Only Equations 2 and 3 are referenced. External scientific validation was explicitly excluded. **Decision:** Accepted the bounded check; retained equations unchanged and recorded optional notation improvements for author review.

Completed Session: 3 Tables and Figure

Exact submitted prompt: Task 3. Read-only table and figure audit: check both native tables (7x4 and 9x3 including headers), captions, wrapping widths, and the relative figure path. Report concrete layout risks and smallest fixes, without editing files or inventing measured results.

Actual response summary: Confirmed both table dimensions, width arithmetic, captions, and the 2100-by-720-pixel figure at its relative path. No clipping or distortion was reported. Optional suggestions were a wider first column, smaller array stretch if later content expanded, flexible columns for another template, and larger figure footer text for journal downscaling. **Decision:** Retained the readable current layout. These were conditional refinements, not required repairs. Prism's no-box-warning claim for its test does not supersede the independent Overleaf log, which contains an underfull bibliography line.

Completed Session: 4 Bibliography and References

Exact submitted prompt: Task 4. Read-only bibliography and cross-reference audit. Check references.bib keys R01-R20 against all citations; list missing/unused keys and duplicate/unresolved labels. Verify section, figure, table and equation reference categories. Do not add references or claim external metadata verification.

Actual response summary: Exactly 20 keys, each present once and cited; no missing/unused keys, duplicate labels, or unresolved source targets. All four reference categories are present. The unused sec:transparency label on a starred section would be unsuitable for a future numeric reference. **Decision:** Accepted the checks, retained the unused label because it is not referenced, and added no sources. External metadata was not independently verified by this Prism prompt.

Completed Session: 5 Debugging

Exact submitted prompt: Task 5. Read-only debugging guidance for three REAL local conversion issues already fixed before import: blank lines inside equation caused Missing \$ inserted; plain numeric citations required BibTeX; hard-coded figure/table numbers needed dynamic labels. Explain cause and minimal fix for each. Do not claim Prism caused these errors or edit files.

Actual response summary: Blank lines cause invalid paragraph breaks inside mathematics and should be removed; plain citation numbers should become keyed citations with bibliography passes; labels belong after captions and object references should be dynamic. Prism explicitly did not attribute these errors to itself. **Decision:** Accepted as retrospective confirmation of the documented local fixes, not as newly discovered platform failures.

Completed Session: 6 Paragraph Revision

Exact submitted prompt: Task 6. Without editing files, improve the paragraph beginning "Calibration estimates model parameters" in main.tex for clarity. Preserve the three separate activities and all three confounding mechanisms, with no new claims. Return only a proposed Version B and a concise explanation of changes; student approval is still pending.

Actual Version B, verbatim: Model calibration estimates model parameters, data assimilation updates the estimated system state as observations become available, and sensor calibration addresses measurement accuracy. These three activities should be documented separately; otherwise, an apparently successful hydraulic calibration may compensate for sensor bias, incorrect irrigation inputs, or unmodelled water-table influence.

Actual explanation summary: Improved parallel structure, clarified sensor calibration, and tightened the warning while preserving all three activities and confounding mechanisms. **Decision:** The student explicitly selected "Use Prism's revised paragraph" on 27 September 2026. Version C therefore equals Version B, approved without further wording changes.

Revision Follow-Up

Actual outcome: Prism applied the approved paragraph and compiled successfully. Its final export differs from the first export only in that paragraph; the bibliography and figure are byte-identical. The exported main.tex was uploaded unchanged to Overleaf and recompiled. Both downloaded final PDFs have 13 pages and matching normalized page text. No additional manuscript corrections were required.

Exact submitted prompt: Task 7. The student explicitly approved Version B from your Task 6. Replace only the paragraph beginning "Calibration estimates model parameters" in main.tex with that exact Version B. Do not change any other text, equations, tables, references or files. Compile and report the one-paragraph change.

This follow-up implements the student's choice after the initial unchanged-export test. Original first-run PDFs, ZIP, and compiler logs are preserved separately. The first six prompts completed on 26 September; the approved revision follow-up occurred on 27 September. The initial generic chat response is excluded from the meaningful-prompt count.